Latent Domain-Specific Prompt-Driven SAM via Conditional Diffusion Refinement for Post-operative Glioma Segmentation

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***Abstract*— Post-operative glioma segmentation in AI-enhanced diagnostic systems presents unique challenges compared to pre-operative tasks, primarily due to the difficulty in distinguishing scattered minute residuals and treatment-induced artifacts amidst severe cross-center heterogeneity. To address these challenges, we propose DPD-SAM, a unified domain-informed refinement framework. The core innovation lies in a coherent pipeline where intrinsic imaging patterns are first distilled offline into pseudo-domain priors via unsupervised clustering, which then drive a jointly optimized prompt-adaptation and diffusion-refinement process. These priors are dynamically synthesized into low-rank prompts to equip the foundation model with domain-specific awareness. Subsequently, the initial masks serve as geometric conditions for a generative diffusion process, which iteratively recovers fine anatomical details and corrects boundary defects in a closed-loop manner. Extensive experiments on BraTS 2024 and three other multi-center datasets demonstrate that DPD-SAM achieves a Dice Similarity Coefficient of 86.8%, consistently outperforming strong baselines, which highlights its superior cross-domain adaptability and precise segmentation of minute residual lesions. This framework offers a promising step toward automating post-operative glioma segmentation and offers new insights for the robust adaptation of vision foundation models to heterogeneous medical tasks.**

***Index Terms*— Post-operative Glioma Segmentation, Segment Anything Model, Latent Domain Mining, Prompt Learning, Diffusion Models.**

I. Introduction

As a critical component of intelligent computer-aided diagnosis (CAD) systems, Glioma is one of the most invasive and common primary brain tumors. Accurate segmentation of residual tumors and surgical cavities in postoperative magnetic resonance imaging (MRI) is essential for treatment planning, response assessment, and prognosis [1]. According to international guidelines such as the Response Assessment in Neuro-Oncology (RANO) criteria [2], the volume of residual tumors, the scope of edema, and the morphology of surgical cavities in postoperative magnetic resonance imaging (MRI) directly determine the formulation of subsequent radiotherapy and chemotherapy regimens, and are significantly correlated with patient recurrence risk and overall survival [3]. However, postoperative brain images exhibit complex pathological characteristics. Specifically, residual tumors often exhibit a diffuse distribution, while surgical disruption of the blood-brain barrier (BBB) causes confounding non-tumorous patchy enhancement on T1-weighted sequences [4]. Moreover, inflammatory responses and edema induced by concurrent chemoradiotherapy further obscure anatomical boundaries. As a result, the intensity contrast between surgical cavities and residual tumors is minimal, and local brain tissue structures undergo severe morphological changes due to surgical trauma. These complex factors render traditional manual contouring of tumors not only extremely time-consuming but also prone to inter-observer variability, making it difficult to meet the standardized requirements of real-time intelligent clinical workflows and multi-center research.

The advancement of deep learning techniques has propelled the development of automated brain MRI segmentation. Seminal architectures, such as the U-Net series [5] and Transformer-based models [6], have achieved promising performance in preoperative glioma segmentation on public benchmarks like BraTS [7]. Nevertheless, these methods encounter significant bottlenecks when applied to postoperative scenarios, primarily due to the scarcity of annotated post-operative datasets and the structural heterogeneity induced by varying surgical approaches. Consequently, compared to preoperative segmentation tasks, the literature on this problem remains relatively scarce, with over 98% of published studies focusing exclusively on preoperative imaging.

The emergence of Foundation Models, particularly the Segment Anything Model (SAM) [8], offers a potential solution to data scarcity via its powerful zero-shot generalization capabilities. Despite its success in natural images and general medical 3D adaptation (e.g., SAM-Med3D [9]), directly applying SAM to postoperative glioma segmentation remains problematic. Standard SAM relies on sparse geometric

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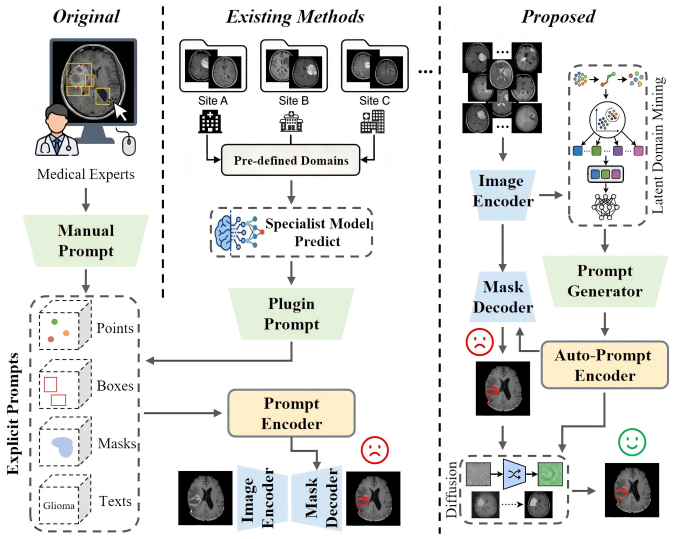


Fig. 1. Comparison of SAM-based segmentation pipelines. (**Left**) Traditional SAM requires manual prompts from medical experts, limiting clinical usability and reproducibility. (**Middle**) Existing SAM-based methods rely on pre-defined domains and complex specialist models for prompt generation, failing to handle multi-center domain shift and postoperative artifacts. (**Right**) Our method introduces unsupervised latent domain discovery to automatically generate adaptive domain-specific prompts for driving SAM, followed by conditional diffusion-based mask refinement, achieving robust and precise segmentation without expert intervention.

prompts (points/boxes) which carry no information about the underlying imaging style [10]. In multi-center scenarios, SAM struggles to adapt to scanner-specific noise patterns or surgical artifacts without explicit domain guidance [11]. Furthermore, the mask decoder of SAM is designed for general object segmentation and often yields over-smoothed boundaries. It lacks the generative capacity to reconstruct the intricate, jagged margins of resection cavities, which are crucial for precise radiotherapy planning. While Diffusion Models have shown promise in generative refinement [12], integrating them into a unified segmentation pipeline with specific anatomical constraints remains an open challenge.

To bridge these gaps, as illustrated in Fig. 1 (Right), we propose DPD-SAM, a unified framework integrating Latent Domain Mining, Dynamic Prompt Adaptation, and Conditional Diffusion Refinement. We clarify that the latent domain mining is performed offline, using K-means++ clustering on features from a frozen SAM encoder to produce pseudo-domain priors, while the dynamic prompt adaptation and the conditional diffusion refinement are subsequently optimized jointly under a single training objective. To the best of our knowledge, DPD-SAM is the first framework that integrates (i) unsupervised latent domain discovery directly from the global semantic space of a frozen SAM encoder, (ii) dynamic low-rank domain-specific prompt generation with affinity-based interpolation, and (iii) ternary-conditioned diffusion refinement into a single coherent pipeline. Unlike prior SAM adaptations [8], [9] that rely on discrete manual prompts or pre-defined center labels, our method introduces structural innovations to jointly resolve severe multi-center heterogeneity and boundary precision issues without expert intervention.

Specifically, unlike conventional domain adaptation methods that require explicit source-domain supervision [13], [14], we introduce an Unsupervised Latent Domain Mining mechanism. It leverages the global semantic space of the SAM encoder to automatically identify latent imaging patterns and assign pseudo-domain labels. To efficiently adapt SAM to these discovered domains, we design a Dynamic Low-Rank Domain Prompt Adaptation strategy. In our framework, a prompt is not a standard text or raw image input, but a set of continuous, learnable low-rank visual tokens serving as a domain-specific prefix. We term this module a generator because it dynamically synthesizes these instance-specific prompts by interpolating learnable low-rank bases [15] conditioned on the latent domain affinity, thereby enabling robust initial localization. Finally, to address the boundary precision issue, we propose a Conditional Diffusion Refinement (CDR) module. By reformulating refinement as a generative process guided by a ternary conditional system (anatomical, spatial, and domain conditions), CDR effectively repairs segmentation defects and recovers micro-lesions missed by the initial pass. Furthermore, we conduct comprehensive evaluations on BraTS 2024, the largest publicly available post-operative glioma dataset comprising 1350 multi-center cases, alongside three additional heterogeneous cohorts to validate real-world clinical robustness. In summary, our main contributions are as follows:

1) We propose an unsupervised latent domain mining strategy. By leveraging the global semantic space, this module automatically captures multi-center heterogeneity without explicit labels or hand-crafted style features, providing semantically rich pseudo-domains for downstream adaptation.
2) We design a dynamic prompt adaptation mechanism. Through parameter-efficient low-rank decomposition and an affinity-based Adapter, this method enables SAM to synthesize instance-specific prompts, achieving effective cross-domain adaptation and generating robust initial segmentation masks that serve as reliable geometric priors for subsequent refinement.
3) We introduce a Domain-Constrained Conditional Diffusion Refinement module to perfect the segmentation details. Building upon the initial masks, this module forms a generative task conditioned on anatomical, spatial, and domain priors, it effectively repairs postoperative boundary defects, achieving state-of-the-art performance and superior generalization on clinical data.

II. Related Work

A. Postoperative Glioma Segmentation

Automatic segmentation of postoperative gliomas is critical for treatment planning but remains computationally challenging due to surgical cavity interference, residual tumor heterogeneity, and blurred anatomical boundaries [16]. Early approaches were predominantly based on Convolutional Neural Networks (CNNs), particularly the U-Net architecture and its variants [5]. Notable frameworks such as nnU-Net optimized hyperparameter configuration to accommodate medical imaging characteristics, establishing a strong baseline [17].

To enhance feature discrimination, subsequent studies introduced attention mechanisms to explicitly focus on tumor-enhancing regions and edema boundaries [18]. However, these discriminative models often fail to capture the specific tissue patterns of postoperative scenarios, struggling to distinguish between fibrous surgical cavities and residual tumor tissue. While recent multi-modal fusion strategies have improved robustness [19], they frequently neglect variations in multi-center data distributions, leading to performance degradation when applied to heterogeneous datasets from different institutions.

B. SAM-based Medical Image Segmentation

Similar to Large Language Models (LLMs) like GPT-4, Vision Foundation Models (VFMs), particularly the Segment Anything Model (SAM), have demonstrated strong zero-shot generalization capabilities [8]. To adapt SAM to the medical domain, researchers have proposed strategies such as MedSAM, which fine-tunes SAM on large-scale multi-modal datasets using bounding box prompts [11], and MSA, which incorporates adapter modules into the ViT encoder to capture task-specific information [20]. Recent works, such as AutoSAM [21] and SurgicalSAM [22], have introduced automated prompting mechanisms to reduce manual intervention. Despite these advancements, existing methods predominantly rely on sparse geometric prompts or generic image embeddings, failing to fully exploit the latent pathological patterns of postoperative tissue. Furthermore, fixed prompt designs are ill-suited for multi-center data heterogeneity, as they lack mechanisms to adjust to varying scanner protocols or domain shifts. Consequently, developing domain-specific prompt generation strategies that can dynamically adapt to data distribution shifts remains a critical gap in SAM-based glioma segmentation.

C. Diffusion Models for Medical Image Segmentation

Diffusion Probabilistic Models (DPMs) have emerged as a powerful paradigm in medical imaging, achieving remarkable success in image generation, segmentation, and classification [23]. Early applications primarily focused on improving generalization by generating synthetic samples to augment training data. However, these techniques often necessitate computationally expensive fine-tuning or require access to target domain data. More recently, the focus has shifted towards leveraging the semantic information encoded in diffusion features for segmentation refinement. These approaches aim to restore blurred edges and missing details from coarse foundation model predictions through iterative denoising [24], [25]. Nevertheless, existing refinement methods rarely integrate latent domain constraints or provide targeted attention to complex regions such as surgical cavity margins. In addition, GAN-based alternatives risk mode collapse and struggle with structural consistency [26]. Unlike standard approaches that rely solely on image conditioning, this study leverages the latent space of diffusion models to efficiently capture class- and domain-specific information, using a conditional generative process to progressively refine segmentation boundaries and recover micro-lesions.

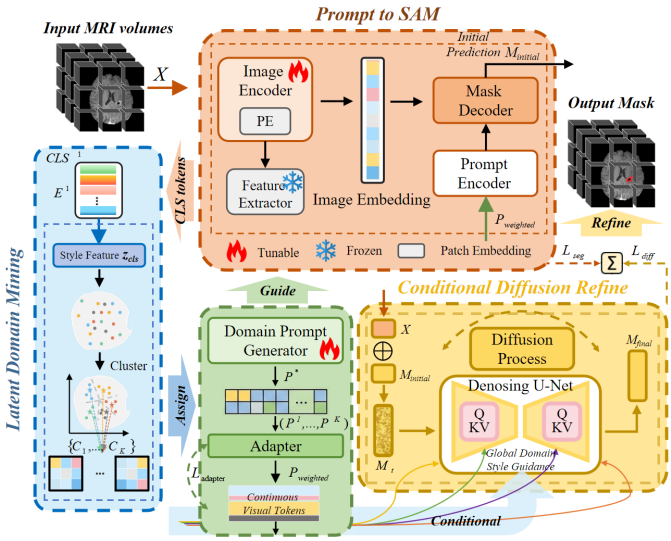


Fig. 2. Overall architecture of the proposed DPD-SAM framework. Multi-center MRI volumes undergo unsupervised Latent Domain Mining to extract domain-specific style features. A low-rank Domain Prompt Generator dynamically synthesizes weighted domain-specific prompts acting as continuous visual tokens to guide SAM, yielding initial segmentation masks. Finally, the Conditional Diffusion Refinement module performs iterative denoising via a modified U-Net, conditioned on the initial masks and domain features, to produce high-precision final segmentation masks.

III. Method

A. Overview

Fig. 2 illustrates the overall architecture of the proposed DPD-SAM, a unified framework for postoperative glioma segmentation that integrates latent domain mining, domain-specific prompt-driven SAM, and conditional diffusion refinement. Taking post-operative 3D MRI volumes as input, the framework operates through three synergistic stages: First, an unsupervised Latent Domain Mining module extracts domain-specific style features to model multi-source heterogeneity and assigns pseudo-domain labels. Second, a Low-Rank Domain Prompt Generator dynamically synthesizes weighted prompts based on these priors to guide SAM in generating initial segmentation masks with geometric constraints. Finally, a Conditional Diffusion Refinement module utilizes these initial masks and domain features to iteratively refine boundary details and repair segmentation defects. To avoid ambiguity regarding the optimization scheme, we explicitly delineate which components are computed offline and which are optimized jointly: (i) the pre-trained SAM backbone is adopted as the feature extractor and provides the initialization of the model; (ii) the Latent Domain Mining, namely the K-means++ clustering and the resulting pseudo-domain labels, is computed offline prior to training and is not updated by gradients; and (iii) the remaining components, including the Domain Prompt Generator, the Prompt Adapter, the SAM mask decoder, and the Conditional Diffusion Refinement network, are *optimized jointly* under the composite objective in Eq. (12). Thus, DPD-SAM is best characterized as an offline latent-domain discovery stage followed by a jointly optimized prompt-adaptation-and-refinement stage, rather than a strictly single-

pass end-to-end optimization. By progressively collaborating across these modules, DPD-SAM effectively addresses the challenges of cross-center data heterogeneity and complex post-operative artifacts, ensuring high-precision segmentation of surgical cavities and residual tumors.

B. Unsupervised Latent Domain Mining

In multi-center postoperative glioma segmentation, data distribution shifts are exacerbated by a combination of hardware discrepancies, such as varying imaging protocols, and complex surgical alterations including resection cavity morphology and drainage artifacts. Different from conventional domain generalization methods that require explicit center labels or hand-crafted style features [14], our Unsupervised Latent Domain Mining (LDM) module leverages a learnable global token built upon the frozen SAM encoder to discover pseudo-domains in a completely unsupervised manner. This provides semantically rich priors that effectively decouple hardware-induced variations from pathological semantics, which previous representation learning techniques have not exploited for post-operative scenarios.

Instead of manual feature engineering, we leverage the robust representation capability of the pre-trained SAM image encoder. We hypothesize that the global embedding space of the Vision Transformer backbone naturally encodes distinct domain signatures ranging from acquisition-specific intensity distributions to structural texture patterns. Given an input MRI volume $X \in \mathbb{R}^{C\times D\times H\times W}$, where $C = 1$ for the T1ce sequence adopted in this work, we feed it into the frozen SAM encoder $\mathcal{F}$ to project the data into a latent feature space. It should be noted that the standard SAM-Med 3D encoder produces only dense patch embeddings and does *not* contain a built-in classification token. To obtain a compact global descriptor, we therefore prepend an additional learnable [CLS] token to the patch-embedding sequence, following the standard ViT design. Propagated through the Transformer blocks, this token aggregates global contextual information via self-attention, and its final-layer output is taken as the global feature vector z_{cls}:

$$z_{cls} = \mathcal{F}(X)_{[cls]} \in \mathbb{R}^L \tag{1}$$

where L denotes the embedding dimension. This learnable global descriptor serves as a compact summary of the volume, encoding a coupled representation of global imaging styles, such as contrast distribution, and coarse anatomical semantics like cavity extent. Since domain shifts in medical imaging are often intertwined with specific pathological biases, clustering on z_{cls} allows the model to identify latent domains that reflect both acquisition conditions and clinical complexity.

Upon obtaining the set of global representations $\mathcal{Z} = \{z^1_{cls}, z^2_{cls}, \ldots, z^N_{cls}\}$ for all training samples, we employ the K-means++ clustering algorithm to partition the latent space into K domains. The cluster centroids $\mathcal{C} = \{C_1, \ldots, C_K\}$ are optimized by minimizing the intra-cluster distance:

$$\arg\min_{\mathcal{C}} \sum_{k=1}^{K} \sum_{z_i \in \mathcal{C}_k} \|z_i - C_K\|_2^2 \tag{2}$$

During inference, each input sample is assigned a pseudo-domain label by calculating the affinity between its global token z_{cls} and each centroid. This foundation-model-driven clustering effectively decouples multi-center heterogeneity, providing a semantically rich foundation for the subsequent domain-aware prompt generation.

C. Dynamic Domain Prompt Generation and Adaptation

In postoperative glioma segmentation, residual lesions are frequently confounded with resection cavities, edema, and treatment-induced enhancement foci, significantly diminishing the discriminative power of standard SAM embeddings. Furthermore, relying solely on discrete pseudo-domain labels hinders effective knowledge sharing across the continuous spectrum of multi-center feature distributions. To address this, we introduce a Low-Rank Domain Prompt Generator (DPG) coupled with a lightweight Prompt Adapter. This module dynamically constructs instance-specific prompts by retrieving and interpolating domain knowledge from a learnable basis.

To effectively decouple anatomical commonalities from imaging specificities, we depart from the paradigm of standard discrete geometric prompts such as points and boxes. Instead, our DPG introduces a parameter-efficient, low-rank decomposition mechanism to synthesize continuous visual embeddings. Specifically, the prompt basis $P^k \in \mathbb{R}^{N\times D}$ for the k-th latent domain is formulated as the coupling of a globally shared template and a domain-specific low-rank modulation:

$$P^k = P^* \odot (U_k \cdot V_k^T) \tag{3}$$

where $P^* \in \mathbb{R}^{N\times D}$ is a randomly initialized shared base prompt designed to capture universal glioma anatomy, such as common tissue structural patterns; $\odot$ denotes the Hadamard product; and $U_k \in \mathbb{R}^{N\times 1}$ and $V_k \in \mathbb{R}^{D\times 1}$ are learnable low-rank vectors specific to domain k. This formulation allows the model to reconstruct complex domain-specific style variations—ranging from scanner-induced noise textures to surgical margin artifacts—via the rank-1 term $U_k \cdot V_k^T$, while maintaining a minimal parameter footprint. Rank 1 suffices because inter-domain differences are dominated by global acquisition-style shifts, which are intrinsically low-rank, and it is the most parameter-efficient choice; larger ranks add parameters without consistent gains.

To accommodate ambiguous samples positioned at the boundaries of latent domains, we introduce a dynamic weighting mechanism. A lightweight Adapter network $\mathcal{A}$, comprising a two-layer MLP and a Softmax head, maps the global semantic feature z_{cls} (derived in Section III.B) to a domain affinity distribution w:

$$w = \text{Softmax}(\text{MLP}(z_{cls})) \in \mathbb{R}^K \tag{4}$$

Consequently, the final domain-aware prompt $P_{weighted}$ is synthesized as a linear combination of the domain bases, weighted by the sample's specific affinity:

$$P_{weighted} = \sum_{k=1}^{K} w_k \cdot P^k \tag{5}$$

We term this module a Generator because the prompt is not statically retrieved from a fixed lookup table, but dynamically synthesized on-the-fly for each input instance. The generated $P_{weighted}$ is injected into the SAM mask decoder as a continuous visual prefix, concatenated with sparse prompt embeddings E_{sparse} to form the augmented sequence $E_{input} = [P_{weighted}; E_{sparse}]$. This design enables the frozen SAM image encoder to receive tailored, domain-aware guidance without requiring any parameter updates to the backbone itself. To ensure the Adapter learns semantically meaningful assignments, we supervise the weights w using the pseudo-domain labels y_{pseudo}:

$$\mathcal{L}_{adapter} = \text{CrossEntropy}(w, y_{pseudo}) \tag{6}$$

Additionally, we apply Low-Rank Adaptation to the SAM mask decoder. This ensures that the pre-trained decoder can effectively align with the newly introduced domain-specific prompt signals. Through this prompt-driven adaptation, DPD-SAM effectively mitigates interference from postoperative artifacts, providing robust initial geometric constraints for subsequent diffusion refinement.

D. Domain-Constrained Conditional Diffusion Refinement

Although the domain-specific prompt-driven SAM effectively locates core regions of postoperative lesions, challenges such as surgical margin fibrosis, grayscale overlap between residual tumors and post-treatment enhancement foci, and multi-center imaging variations prevent the coarse mask M_{initial} from attaining pixel-level clinical precision. To address this, we propose a Conditional Diffusion Refinement (CDR) module. Diffusion is highly effective in our framework because we reformulate the refinement task as a generative process rather than a purely discriminative one. Rather than using standard diffusion for generic image synthesis or data augmentation, we mathematically reformulate the reverse denoising process as a ternary-conditional system. While the SAM image encoder remains frozen to preserve its zero-shot generalization, the CDR explicitly constrains the denoising network with anatomical (X), spatial (M_{initial}), and domain (z_{cls}) priors. This generative formulation uniquely forces the model to recover fine-grained anatomical details and jagged resection cavity margins that discriminative approaches inherently miss.

We represent the ground-truth segmentation label as a continuous distribution $M_0 \in \mathbb{R}^{D\times H\times W}$, i.e., a single-channel 3D mask, and all diffusion operations are performed on full 3D volumes via 3D convolutions. During the forward process, Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is gradually added to M_0 according to a fixed variance schedule β_t. The noisy mask M_t at time step t is defined as:

$$M_t = \sqrt{\bar{\alpha}_t} M_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \tag{7}$$

where $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. This process transforms clear anatomical boundaries into a stochastic distribution, creating a challenging denoising objective that compels the model to learn structural recovery.

The core of our method lies in the reverse denoising process, where we aim to predict the added noise based on a multi-source conditional signal. As mentioned, we design a ternary conditional system: (1) *Anatomical Condition*: the raw post-operative MRI volume X; (2) *Spatial Condition*: the initial mask $M_{initial}$ generated by SAM; and (3) *Domain Condition*: the global latent feature z_{cls}. To integrate these conditions, the denoising network ϵ_θ is implemented as a modified 4-layer 3D U-Net with the following key architectural differences from the original U-Net: (i) each encoder/decoder stage uses two $3 \times 3 \times 3$ convolutional layers with instance normalization and LeakyReLU activation; (ii) channel-wise attention is introduced in the bottleneck layer to enhance feature recalibration; (iii) cross-attention modules are embedded in the three deepest decoder layers to inject the global domain feature z_{cls} as a style condition, serving as the key and value to guide the spatial query features; and (iv) skip connections are preserved with additional $1 \times 1 \times 1$ convolutions for feature fusion. Furthermore, to preserve strict spatial alignment, the input layer receives a multi-channel tensor formed by concatenating X and $M_{initial}$. Since this work adopts the single-modality T1ce sequence with $C = 1$, the spatial input to the denoising network consists of two channels, namely the single-channel anatomical volume X and the single-channel initial mask $M_{initial}$; the domain feature z_{cls} is injected separately through the cross-attention layers rather than as an input channel. We deliberately adopt a relatively shallow 4-layer design: since the CDR only refines an already well-localized initial mask on a low-resolution grid, with global context supplied by the frozen SAM encoder and the domain prior, a deeper denoiser brings little benefit while increasing parameters and latency. These modifications enable the network to effectively perceive fine anatomical details while adaptively compensating for domain-specific artifacts. The noise prediction is formulated as:

$$\hat{\epsilon} = \epsilon_\theta(M_t, t, \text{Concat}(X, M_{initial}), z_{cls}) \tag{8}$$

To ensure the refined mask adheres to both anatomical consistency and latent domain logic, the diffusion refinement loss is defined as:

$$\mathcal{L}_{diff} = \mathbb{E}_{M_0,\epsilon,t}[\|\epsilon - \epsilon_\theta(M_t, t, \text{Concat}(X, M_{initial}), z_{cls})\|^2] \tag{9}$$

During inference, the model starts from pure Gaussian noise M_T and performs iterative reverse sampling. Under the strong geometric guidance of $M_{initial}$ and the style constraints of z_{cls}, the stochastic noise is progressively refined into a high-precision mask M_{final}, helping correct boundary defects and better delineate small residual lesions, as shown in Fig. 5 and Table VII. Besides the noise-prediction loss, the recovered mask is supervised by a Dice+BCE loss given by the $\mathcal{L}_{seg}$ term in Eq. (12), and at test time it is binarized at 0.5 with no post-processing.

E. Objective Functions and Optimization Strategy

To effectively train the jointly optimized stage of the proposed framework, we design a composite loss function system. The total objective is a weighted combination of three

specific losses: segmentation loss, diffusion refinement loss, and adapter supervision loss.

First, to guide the Domain-Specific Prompt-Driven SAM in generating robust initial masks, we employ a supervised segmentation loss $\mathcal{L}_{seg}$. Given the inherent class imbalance between small residual tumors and the background, we formulate this loss as a weighted sum of Dice Loss and Binary Cross-Entropy (BCE) Loss:

$$\begin{aligned}\mathcal{L}_{seg} = \lambda_{dice}\mathcal{L}_{Dice}(M_{initial}, M_{gt}) \\ + \lambda_{bce}\mathcal{L}_{BCE}(M_{initial}, M_{gt}),\end{aligned} \tag{10}$$

where $M_{initial}$ is the predicted mask from SAM, M_{gt} is the ground truth, and $\lambda_{dice}, \lambda_{bce}$ are balancing coefficients. This loss updates the parameters of the Prompt Generator, Adapter, and the SAM decoder.

Second, to govern the Conditional Diffusion Refinement, we utilize the diffusion denoising loss $\mathcal{L}_{diff}$ defined in Eq. (9). This term compels the model to learn the non-linear mapping from coarse geometric constraints to precise anatomical boundaries under the guidance of the global domain feature z_{cls}. This loss specifically optimizes the parameters of the Denoising U-Net.

Third, to ensure the Prompt Adapter learns semantically meaningful domain assignments, we introduce the Adapter Supervision Loss $\mathcal{L}_{adapter}$. Instead of relying on implicit feedback, we explicitly supervise the predicted domain weights w using the pseudo-domain labels y_{pseudo} via cross-entropy:

$$\mathcal{L}_{adapter} = \text{CrossEntropy}(w, y_{pseudo}) \tag{11}$$

The total training objective $\mathcal{L}_{total}$ is expressed as:

$$\mathcal{L}_{total} = \mathcal{L}_{seg} + \alpha\mathcal{L}_{diff} + \beta\mathcal{L}_{adapter} \tag{12}$$

where α and β are hyperparameters balancing the tasks.

To mitigate convergence instability during the early stages, we adopt a two-phase warm-up optimization strategy, whose detailed procedure is summarized in Algorithm 1. Based on the empirical sensitivity analysis, we set the warm-up threshold to $E_{warm} = 50$ as the default configuration for all main experiments.

- **Phase 1 (Warm-up):** We temporarily freeze the diffusion module and utilize $\mathcal{L}_{seg}$ and $\mathcal{L}_{adapter}$ to optimize the Domain Prompt Generator and Adapter. This phase ensures the model learns to generate reliable initial masks and accurate domain weights, establishing a solid geometric foundation.
- **Phase 2 (Joint Refinement):** Once the SAM backbone produces stable coarse masks, we activate the training of the CDR module using $\mathcal{L}_{diff}$. In this phase, the framework functions as a cohesive pipeline, where the generative capabilities of the diffusion model are aligned with the anatomical constraints provided by SAM.

Algorithm 1: Training Procedure of DPD-SAM.

Input: Multi-center MRI Data $\mathcal{D}_{train}$, Total Epochs E, Warm-up Epochs E_{warm}
Require: Pre-trained SAM Image Encoder $\mathcal{F}$, Mask Decoder $\mathcal{M}$, Denoising U-Net ϵ_θ
Initialize Shared prompt base $\mathbf{P}^*$, Low-rank vectors U_k, V_k, Prompt Adapter Network $\mathcal{A}$;
Offline: Unsupervised Latent Domain Mining
Extract global tokens $\mathcal{Z}$ via Eq. (1) and obtain pseudo-labels Y_{pseudo} via Eq. (2);
for $e = 1$ **to** E **do**
  for $(X, M_{gt}, y_{ps}) \sim \mathcal{D}_{train}$ **do**
    Extract image embeddings z_{img} and global token z_{cls} via $\mathcal{F}(X)$;
    Compute domain weights w and weighted prompt $\mathbf{P}_{weighted}$ via Eq. (4) and Eq. (5);
    Generate initial mask $M_{initial} = \mathcal{M}(z_{img}, \mathbf{P}_{weighted})$;
    if $e \leq E_{warm}$ **then**
      // Phase 1: Warm-up
      Calculate $\mathcal{L}_{warmup} = \mathcal{L}_{seg} + \beta\mathcal{L}_{adapter}$ via Eq. (10) and (11);
      Update $\mathbf{P}^*, U_k, V_k, \mathcal{A}$ and $\mathcal{M}$ using $\nabla\mathcal{L}_{warmup}$;
    else
      // Phase 2: Joint Refinement
      Sample $t \sim \text{Uniform}(1, T)$ and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, compute M_t via Eq. (7);
      Predict noise $\hat{\epsilon}$ via Eq. (8) using input Concat$(X, M_{initial})$ and z_{cls};
      Calculate $\mathcal{L}_{total}$ via Eq. (12);
      Update $\mathbf{P}^*, U_k, V_k, \mathcal{A}, \mathcal{M}$ and ϵ_θ using $\nabla\mathcal{L}_{total}$;
    end
  end
end

IV. Experiments

A. Experimental Settings

1) Datasets: To comprehensively evaluate the proposed method, we conducted experiments on three registered 3D MRI datasets sourced from multiple clinical centers. BraTS 2024 [7], released by the Brain Tumor Segmentation Challenge, serves as the primary large-scale benchmark. It is currently the largest expert-annotated MRI dataset for post-treatment glioma, comprising 1350 clinical cases jointly provided by seven academic medical centers. RHUH [27] and ReMIND [28] are two small-sample public datasets, from which we utilized 40 and 30 cases, respectively. Furthermore, an internal dataset consisting of 97 cases was obtained from our local institution. Detailed information regarding these datasets is summarized in Table I.

Ground truth segmentation of residual tumor tissues was

TABLE I
SUMMARY OF DATASETS AND TUMOR CHARACTERISTICS ACROSS DATASETS. THE STR RATE INDICATES THE PERCENTAGE OF CASES WHERE THE RESIDUAL TUMOR VOLUME IS < 5 ML, FOLLOWING RANO CRITERIA.

Datasets	Contributing Sites	STR Rate	Volume(ml)*
BraTS2024	Duke University Univ. of California SF Univ. of Missouri Columbia Univ. of California SD Heidelberg Univ. Hospital Univ. of Michigan Indiana University	2.8%	41.64 (62.14)
RHUH	Rio Hortega Univ. Hospital	7.5%	26.55 (37.12)
ReMIND	Brigham & Women's Hospital	53.3%	4.69 (12.37)
In-house	Our hospital	71%	2.18 (5.45)

* Volumes are expressed as Median (Interquartile Range).

TABLE II
DETAILED ARCHITECTURAL SPECIFICATIONS OF THE DENOISING NETWORK ϵ_θ.

Component	Specifications	Output Size ($C \times D \times H \times W$)
Input Layer	Concat(X, M_{initial})	$2 \times 128 \times 128 \times 128$
Encoder Level 1	$3 \times 3 \times 3$ Conv, Stride 1, Channel 64	$64 \times 128 \times 128 \times 128$
Encoder Level 2	$3 \times 3 \times 3$ Conv, Stride 2, Channel 128	$128 \times 64 \times 64 \times 64$
Encoder Level 3	$3 \times 3 \times 3$ Conv, Stride 2, Channel 256	$256 \times 32 \times 32 \times 32$
Encoder Level 4	$3 \times 3 \times 3$ Conv, Stride 2, Channel 512	$512 \times 16 \times 16 \times 16$
Bottleneck	Channel-wise Attention	$512 \times 16 \times 16 \times 16$
Decoder Level 3	Cross-Attention (Query: Feat, Key/Val: z_{cls})	$256 \times 32 \times 32 \times 32$
Decoder Level 2	Cross-Attention (Query: Feat, Key/Val: z_{cls})	$128 \times 64 \times 64 \times 64$
Decoder Level 1	Cross-Attention (Query: Feat, Key/Val: z_{cls})	$64 \times 128 \times 128 \times 128$
Skip Connections	$1 \times 1 \times 1$ Conv + Addition	-
Normalization	Instance Normalization	-
Activation	LeakyReLU ($\alpha = 0.01$)	-

performed on T1-weighted contrast-enhanced (T1ce) MRI scans by professionally trained annotators and subsequently refined by neuroradiologists and neurosurgical experts. For the in-house cohort, the use of the retrospectively collected, de-identified data was approved by the institutional ethics committee under approval No. HS2021-504. Each case was independently annotated by trained neurosurgeons and reviewed by senior experts to establish a consensus reference standard. Following international guidelines for Extent of Resection (EOR) [2], we categorized cases with a residual tumor volume of less than 5 ml as clinically satisfactory Subtotal Resection (STR). The post-operative cohort was randomly partitioned into training and testing sets at a ratio of 8:2. Approximately 20% of the training set was reserved as a validation set. Data splitting was performed at the patient level to ensure that all scans from the same individual remained within the same cohort. The training set was used exclusively for model optimization, and all performance metrics were evaluated on the independent testing set. In addition, the RHUH, ReMIND, and in-house cohorts are kept entirely separate from training and serve as independent external test sets for assessing cross-center generalization.

2) Implementation Details: All experiments were conducted using the PyTorch framework on a Linux-based workstation equipped with two MetaX C500 GPUs (65GB VRAM each). The SAM-Med 3D model was employed as the backbone architecture. All MRI volumes underwent a standardized preprocessing pipeline, including intensity normalization, bias field correction, and skull stripping. Subsequently, data were resized to a consistent input dimension of $128 \times 128 \times 128$. To improve the model's generalization ability across multi-center domains, we applied extensive data augmentation techniques via TorchIO during training, including random affine transformations (rotation up to $15°$, translation up to 10%), random elastic deformations, and random Gaussian noise injection.

Network parameters were optimized using the AdamW optimizer with an initial learning rate of 4×10^{-4} and a weight decay of 0.1. The model was trained for 200 epochs with a batch size of 2. For the conditional diffusion refinement, the number of forward training timesteps T was set to 1000, and we utilized DDIM sampling with $S = 10$ steps during inference to accelerate the process, where T denotes the training diffusion steps and S the inference sampling steps

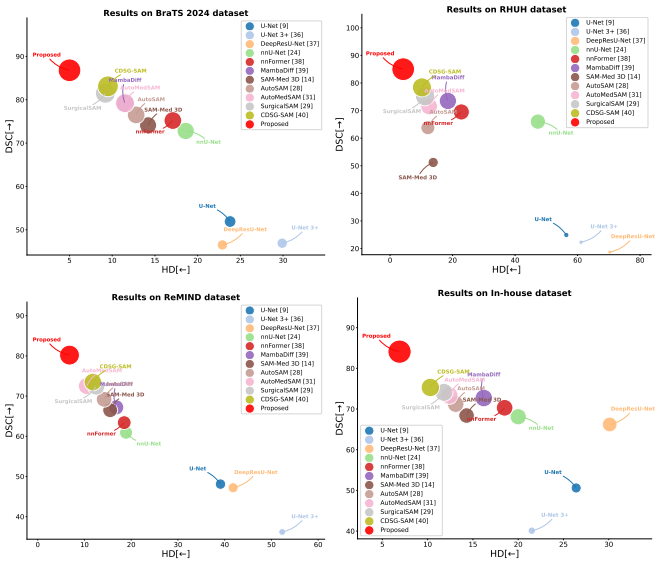


Fig. 3. Visualization of comparative performance on BraTS 2024, RHUH, ReMIND, and In-house datasets. The X-axis corresponds to the HD95 (lower is better), while the Y-axis corresponds to the DSC (higher is better). Additionally, the bubble size is proportional to the IoU (larger is better). The proposed method achieves the best trade-off and consistently outperforms baselines across all datasets.

throughout this paper. Furthermore, the detailed architectural specifications and hyperparameters of the denoising network ϵ_θ are summarized in Table II to ensure full reproducibility. To quantitatively assess the segmentation performance, we utilized the Dice Similarity Coefficient (DSC), Intersection over Union (IoU), and 95% Hausdorff Distance (HD95). All results are averaged over five independent runs that share the same patient-level split but differ in random seed, and are reported as mean $\pm$ standard deviation. The improvements of DPD-SAM over the strongest baseline are statistically significant on all datasets under a paired t-test at $p < 0.001$, with Cohen's d above 5.

B. Results

1) Comparison Methods: To verify the efficacy of the proposed method, we conducted a comparative analysis against several state-of-the-art medical image segmentation models. The comparison suite includes: (1) **Specialist Models**: Established CNN-based architectures (U-Net [5], U-Net 3+ [29], DeepResU-Net [30]), the adaptive framework nnU-Net [17], and the transformer-based nnFormer [31], as well as the recent diffusion-based MambaDiff [32]. (2) **SAM-Based Models**: Specialized adaptations of the Segment Anything Model for medical tasks, including SAM-Med 3D [9], AutoSAM [21], AutoMedSAM [24], SurgicalSAM [22], and CDSG-SAM [33]. To ensure the reliability of the experimental results, all methods were evaluated under identical experimental conditions. Specifically, all models share the same data splits, single-modality T1ce input, preprocessing/augmentation, resolution, and training budget, and each uses its official implementation; SAM-based baselines use automatically generated prompts under a unified protocol without any manual boxes or points. The backbone-agnostic

TABLE III
QUANTITATIVE COMPARISON OF SEGMENTATION PERFORMANCE ACROSS FOUR DIVERSE DATASETS. THE RESULTS ARE PRESENTED AS **MEAN** ± **STD** DERIVED FROM 5 INDEPENDENT RUNS. **BOLD** INDICATES THE BEST PERFORMANCE, AND UNDERLINED INDICATES THE SECOND-BEST RESULTS. DSC: DICE SIMILARITY COEFFICIENT (%); IOU: INTERSECTION OVER UNION (%); HD: 95% HAUSDORFF DISTANCE (MM).

Type	Methods	BraTS 2024			RHUH			ReMIND			In-house		
		DSC ↑	IoU ↑	HD ↓	DSC ↑	IoU ↑	HD ↓	DSC ↑	IoU ↑	HD ↓	DSC ↑	IoU ↑	HD ↓
Specialist	U-Net [5]	51.9±2.1	40.1±1.8	23.8±3.5	24.9±3.2	17.7±2.5	56.4±8.1	48.1±2.4	35.6±2.1	39.1±4.2	50.6±2.4	33.9±2.0	26.4±3.1
	U-Net 3+ [29]	46.9±2.5	35.3±2.2	29.9±4.1	22.3±3.0	14.4±2.8	61.1±9.2	36.2±3.1	23.3±2.5	52.3±5.5	40.1±2.9	24.9±2.2	21.5±2.6
	DeepResU-Net [30]	46.5±2.0	35.2±1.9	22.9±3.0	18.7±2.5	12.9±2.1	70.3±8.5	47.2±2.2	34.4±1.8	41.8±4.0	66.2±1.6	49.5±1.4	30.1±3.3
	nnU-Net [17]	72.8±1.2	58.9±1.1	18.6±1.5	66.0±1.5	51.4±1.3	47.3±5.2	60.9±1.8	44.6±1.5	18.9±2.1	68.1±1.2	54.6±1.1	20.0±1.9
	nnFormer [31]	75.2±1.1	60.3±1.0	17.1±1.4	69.5±1.3	54.2±1.2	22.8±3.5	63.4±1.5	47.1±1.4	18.5±1.9	70.3±1.3	55.9±1.2	18.5±1.6
	MambaDiff [32]	79.2±0.9	65.5±0.8	11.5±0.5	73.5±1.1	58.2±1.0	18.5±2.8	67.2±1.4	51.5±1.2	16.8±1.6	72.7±1.1	58.8±1.0	16.2±1.5
SAM-Based	SAM-Med 3D [9]	74.1±1.5	58.2±1.3	14.2±1.2	51.2±2.1	34.5±1.8	13.8±2.0	66.5±1.6	50.8±1.4	15.5±2.2	68.4±1.4	52.2±1.3	14.3±1.6
	AutoSAM [21]	76.5±1.3	60.4±1.2	12.8±1.1	63.8±1.5	47.2±1.3	12.1±1.8	69.2±1.4	53.5±1.3	14.2±1.9	71.2±1.2	55.4±1.1	13.1±1.4
	AutoMedSAM [24]	79.2±1.1	63.8±1.0	11.5±1.0	71.5±1.2	56.4±1.1	12.5±1.5	72.5±1.3	57.8±1.2	10.5±1.6	73.3±1.1	58.7±1.0	12.4±1.2
	SurgicalSAM [22]	81.5±1.0	67.8±0.9	9.2±0.9	75.2±1.1	64.8±1.0	11.2±1.4	72.4±1.2	58.8±1.1	12.5±1.5	74.1±1.0	60.2±0.9	11.8±1.1
	CDSG-SAM [33]	83.1±0.8	70.5±0.7	9.5±0.6	78.4±1.0	64.2±0.9	10.2±1.2	73.5±1.1	59.9±1.0	11.8±1.2	75.3±0.9	61.7±0.8	10.3±1.0
	Proposed	**86.8**±0.6	**77.0**±0.5	**5.0**±0.8	**85.0**±0.8	**76.2**±0.7	**4.2**±1.0	**80.2**±0.9	**67.5**±0.8	**6.8**±1.1	**84.1**±0.8	**77.8**±0.7	**6.9**±0.9

TABLE IV
SEGMENTATION PERFORMANCE ON BRATS 2024 DATASET USING DIFFERENT FOUNDATION MODEL BACKBONES.

Backbone Model	Backbone Only		+ Proposed Modules		Absolute Improvement	
	DSC↑	HD↓	DSC↑	HD↓	ΔDSC↑	ΔHD↓
SAM (ViT-B) [8]	68.4	16.9	82.1	7.8	+13.7	-9.1
MedSAM [11]	72.6	14.5	84.3	6.5	+11.7	-8.0
SAM-Med 3D	74.1	14.2	**86.8**	**5.0**	+12.7	-9.2

study reported in Table IV together with the ablation reported in Table V further confirm that the gains stem from the proposed modules rather than from a particular backbone or training setup.

2) Quantitative Analysis: As summarized in Table III, our method achieved superior performance across multiple datasets, demonstrating the robustness of the framework under multi-center data distributions. On the large-scale BraTS 2024 dataset, DPD-SAM achieves a DSC of 86.8%, surpassing the strongest specialist model by 7.6% and the best prior SAM-based method by 3.7%. This significant performance leap is primarily attributed to the effective extraction of heterogeneous style features by the LDM module and the utilization of powerful semantic priors from the SAM foundation model. Notably, our model exhibits exceptional generalization advantages on smaller, heterogeneous cohorts. More importantly, DPD-SAM demonstrates superior cross-domain generalization on the smaller, more challenging datasets. For instance, on the challenging In-house dataset, DPD-SAM attains 84.1% DSC and 6.9 mm HD95.

Furthermore, to verify that the improvement does not merely depend on the feature extractor, we replaced the frozen SAM-Med 3D backbone with two SAM variants, SAM (ViT-B) [8] and MedSAM [11], keeping all proposed modules (LDM, DPG, CDR) unchanged. As shown in Table IV, our modules bring consistent gains across all three backbones, indicating that the improvement originates from the synergistic design rather than from the feature extractor alone.

3) Qualitative Analysis: We performed a qualitative analysis of the segmentation results, as visualized in Fig. 4. Specialist models, such as nnU-Net and nnFormer, often produce over-smoothed boundaries or exhibit boundary adhesion, incorrectly merging residual tumors with surrounding cavity walls or edema. Similarly, SAM-Med 3D and SurgicalSAM display noticeable over-segmentation or fail to detect tiny lesions due to insufficient adaptation to domain-specific imaging styles and postoperative heterogeneity. In contrast, the LDM module effectively captures multi-center style variations, allowing the Low-Rank DPG to supply SAM with adaptive, instance-specific prompts. This enhances the model's sensitivity to weak pathological signals without manual intervention. Subsequently, the CDR module iteratively restores fine-grained details and corrects boundary defects through its ternary conditioning, yielding crisp delineations even for ribbon-like or fragmented residuals. These visualizations underscore the effectiveness of our coarse-to-fine pipeline in overcoming the core challenges of postoperative glioma segmentation: boundary blurring and the occlusion of small lesions.

C. Ablation Study

To quantitatively evaluate the independent contributions and synergistic effects of our core components, we conducted a comprehensive ablation study covering all logically feasible combinations. The detailed performance matrix is summarized in Table V. The frozen baseline model yielded suboptimal performance due to the inherent domain gap. Introducing a lightweight Adapter provided immediate improvements across all datasets, demonstrating that parameter-efficient fine-tuning is a necessary prerequisite for task alignment. We further emphasize that the gains do not stem merely from added capacity: the DPG uses a parameter-efficient rank-1 decomposition and a lightweight Adapter, so its extra parameters are marginal relative to the frozen backbone. The consistent gains across three frozen backbones reported in Table IV, together with the independent and synergistic contribution of each component reported in Table V, indicate that the improvements arise from

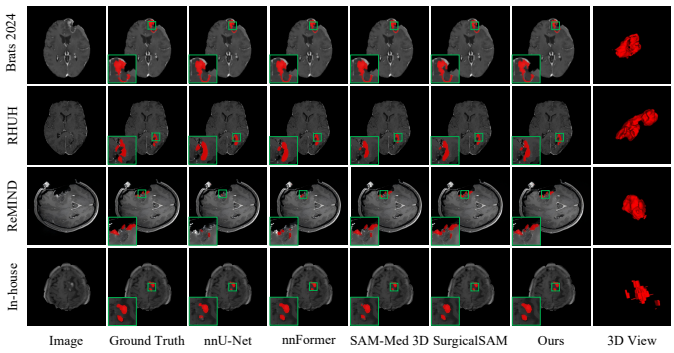


Fig. 4. Visualization of different state-of-the-art methods and our method in four datasets.

TABLE V
ABLATION STUDY OF CRITICAL MODULES IN OUR PROPOSED METHOD. THE EXPANDED MATRIX EVALUATES ALL LOGICALLY FEASIBLE COMBINATIONS TO DEMONSTRATE INDEPENDENT CONTRIBUTIONS AND SYNERGISTIC EFFECTS.

Base	Adapter	LDM	DPG	CDR	BraTS 2024		RHUH		ReMIND		In-house	
					DSC↑	HD↓	DSC↑	HD↓	DSC↑	HD↓	DSC↑	HD↓
✓	-	-	-	-	74.1	14.2	51.2	13.8	66.5	15.5	68.4	14.3
✓	-	-	-	✓	78.2	9.8	56.5	9.2	69.8	11.2	72.3	10.6
✓	✓	-	-	-	78.5	12.8	62.4	12.5	71.2	13.2	73.6	12.8
✓	✓	-	-	✓	82.1	8.5	67.8	8.4	74.5	9.5	76.2	9.4
✓	✓	✓	-	-	81.3	11.5	72.8	11.2	75.6	11.8	78.7	11.3
✓	✓	✓	-	✓	84.2	7.2	77.5	6.8	77.8	8.2	80.8	8.4
✓	✓	✓	✓	-	84.6	9.8	81.5	9.5	78.4	10.2	81.4	10.6
✓	✓	✓	✓	✓	**86.8**	**5.0**	**85.0**	**4.2**	**80.2**	**6.8**	**84.1**	**6.9**

Note: Base = Frozen SAM-Med3D backbone.

the proposed domain-aware mechanisms rather than from a larger parameter budget.

1) Effectiveness of the LDM module: The adapter-tuned baseline SAM model struggles to generalize across diverse centers due to entangled style and semantic features. The integration of the LDM module explicitly models multi-center heterogeneity by clustering samples into latent domains. As shown in Table V, upgrading from the adapter-only configuration to explicitly including LDM yields consistent improvements, particularly on BraTS 2024, where the DSC increased to 81.3%, validating that unsupervised style discovery provides a robust foundation for downstream adaptation.

2) Effectiveness of the DPG module: Building upon the extracted domain priors, the DPG serves as the core engine for cross-center generalization. As shown in Table V, completing the prompt generation pipeline with DPG results in a significant performance leap on the highly heterogeneous RHUH dataset, boosting the DSC to 81.5%. This gain confirms that our dynamic prompting mechanism effectively retrieves domain-specific knowledge to bridge severe domain gaps, significantly outperforming static adaptation strategies.

3) Effectiveness of the CDR module: The CDR module acts as the fine refinement stage, delivering the most pronounced boundary improvements across all datasets. As evidenced by the pairwise comparisons in Table V, the inclusion of CDR consistently improves boundary precision (HD). However, it achieves its optimal synergistic performance only when conditioned on the full domain-aware prompt generation. The visual impact of this refinement is explicitly captured in the spatial pixel-wise error heatmaps shown in Fig. 5. As observed in Fig. 5(c), the model without CDR exhibits significant high-error regions along the jagged cavity margins. In contrast, enabling CDR effectively suppresses these errors, resulting in a clean, low-error heatmap. By reformulating the refinement task as a ternary-conditional generative denoising process, the CDR effectively repairs defects in the initial SAM masks and restores fine-grained details missed due to blurring or minute lesions. The high-error responses concentrate at jagged cavity margins and small low-contrast residuals, which are the regions where the coarse mask is least reliable, and the CDR selectively suppresses these errors while leaving well-segmented areas unchanged.

D. Hyperparameter Sensitivity Analysis

To evaluate the robustness of our framework, we conducted a detailed sensitivity analysis on five key hyperparameters: adapter loss weight β, diffusion loss weight α, cluster number K, the number of DDIM sampling steps S at inference, and warm-up epoch E_{warm}. The experimental results are illustrated in Fig. 6.

1) Impact of Loss Weights (α and β): As shown in Fig. 6(a) and (b), model performance exhibits a consistent bell-shaped distribution relative to the loss weights. Both the adapter supervision weight β and diffusion refinement weight α achieve optimal DSC metrics at 0.1 across all datasets. Setting weights too low fails to provide sufficient gradient guidance for domain adaptation or boundary refinement, respectively. Conversely, excessively high weights overwhelm the primary segmentation objective, leading to optimization instability and degraded performance.

2) Number of Latent Domains (K): Fig. 6(c) analyzes the granularity of unsupervised clustering. Setting $K = 4$ yields the optimal performance. Fewer clusters fail to effectively disentangle complex heterogeneous features, such as distinguishing surgical artifacts from equipment noise. On the other hand, excessive clusters fragment the latent space, reducing the sample size within individual domains and thereby diminishing the generalization capability of the low-rank prompt learning.

3) Diffusion Inference Steps (S): The efficiency of generative refinement is analyzed in Fig. 6(d). The DSC improves rapidly as S increases and reaches its peak at $S = 10$. This indicates that our Conditional Diffusion Refinement (CDR) module possesses high convergence efficiency, requiring only a limited number of denoising steps to recover precise anatomical boundaries. Notably, increasing S beyond 20 leads to slight performance fluctuations, likely due to the over-smoothing of fine details caused by excessive denoising. Therefore, we

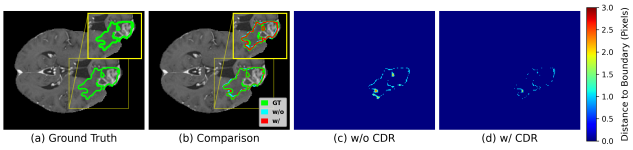


Fig. 5. Comparison of spatial pixel-wise error heatmaps before and after using CDR for boundary refinement.

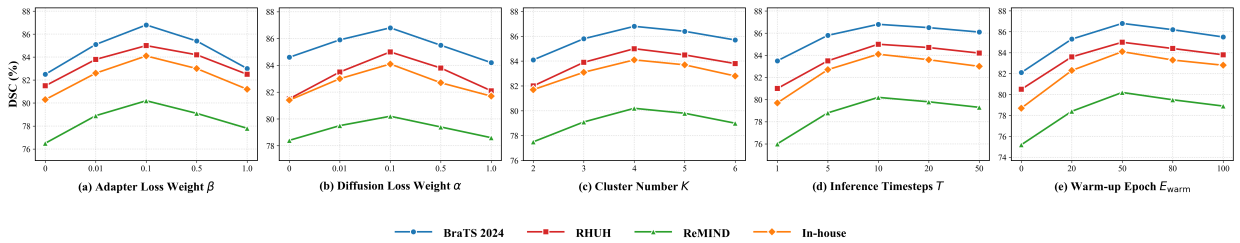


Fig. 6. Hyperparameter sensitivity analysis on four datasets. The subplots illustrate the impact of (a) adapter loss weight β, (b) diffusion loss weight α, (c) number of latent domains K, (d) the number of DDIM sampling steps S, and (e) warm-up epoch E_{warm} on segmentation performance (DSC).

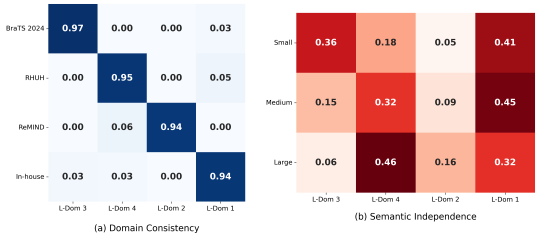


Fig. 7. Effectiveness of latent domain mining: (a) Domain consistency matrix between proxy domains and predicted latent domains, showing strong diagonal alignment. (b) Semantic independence matrix with approximately uniform distribution, confirming good inter-domain separation.

TABLE VI
NMI BETWEEN THE DISCOVERED LATENT DOMAINS AND DIVERSE PATHOLOGICAL/ANATOMICAL DESCRIPTORS VERSUS THE ACQUISITION SITE.

Variable	Category	NMI
Lesion volume (binned)	Pathology	0.0113
Residual/RANO class (STR)	Pathology	0.0017
Lesion shape compactness	Anatomy	0.0216
Lesion location (octant)	Anatomy	0.0122
Intensity mean	Pathology	0.0659
Intensity std	Pathology	0.0117
Acquisition site	Acquisition	0.79

adopt $S = 10$ as the optimal trade-off between segmentation accuracy and clinical inference speed.

4) Warm-up Epoch (E_{warm}): To determine the optimal warm-up epoch E_{warm}, we conducted a sensitivity analysis by varying E_{warm} from 0 to 100 while fixing all other training hyperparameters. As shown in subplot (e) of Fig. 6, performance peaks at $E_{warm} = 50$ across all four datasets. This value allows the Domain Prompt Generator and Adapter to fully converge in Phase 1 before activating the Conditional Diffusion Refinement module in Phase 2, thereby providing stable geometric priors without causing optimization instability. Overall, our method maintains superior performance across a wide range of hyperparameter values, validating the stability of the framework for multi-center clinical applications.

E. Quantitative Analysis of Latent Domains

To validate the effectiveness of the unsupervised latent domain mining, we quantitatively assess the fidelity of the discovered pseudo-domains in capturing multi-center heterogeneity while remaining invariant to semantic content such as lesion characteristics. We employ Normalized Mutual Information (NMI) and confusion matrices for this evaluation.

Fig. 7(a) visualizes the correspondence between the predicted latent domains and the actual site labels. The confusion matrix exhibits strong diagonal dominance, with consistency scores ranging from 0.94 to 0.97. This demonstrates that our token-based clustering reliably recovers intrinsic multi-center patterns without explicit supervision. This near one-to-one correspondence between the discovered clusters and the four data sources gives the latent domains a concrete physical meaning: they primarily encode center-/scanner-specific acquisition styles, which together with the low NMI on pathological and anatomical descriptors reported in Table VI links cluster centroids to known acquisition metadata. Conversely, the semantic-independence matrix in Fig. 7(b) is widely dispersed without a dominant diagonal, confirming that the clustering is driven by acquisition differences rather than lesion scale or complexity.

To provide stronger and more comprehensive evidence of domain–semantic disentanglement, beyond tumor volume alone we further computed the NMI between the discovered latent domains and six pathological/anatomical descriptors covering lesion size, residual-tumor (RANO/STR) category, shape, spatial location, and modality-specific intensity statistics. As summarized in Table VI, all six NMI values are below 0.07, indicating that the discovered domains are essentially independent of lesion morphology, location, and intensity distribution. In sharp contrast, the correspondence between the latent domains and the acquisition site is strong, with a consistency of 0.94–0.97 in Fig. 7(a) and a site-level NMI of 0.79 in Fig. 8. These complementary analyses, consistent with the low semantic-NMI curve in Fig. 8, provide much stronger evidence that the latent domains encode acquisition-/center-specific variations rather than anatomical or pathological differences.

Furthermore, we monitored the clustering stability during a part of training, as illustrated in Fig. 8. The self-consistency NMI between current and previous cluster assignments approaches 1.0 rapidly, stabilizing within the first 5 epochs. Meanwhile, the NMI with respect to the physical domain labels stabilizes at approximately 0.79, while the correlation with semantic labels remains consistently low. These dynamics confirm that the model quickly converges to a stable, domain-correlated, yet semantically independent representation. These

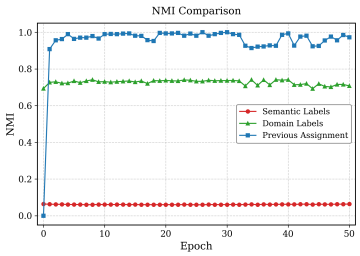


Fig. 8. Normalized mutual information between unsupervised pseudo domain labels and semantic labels, pre-defined domain labels, and previous assignments.

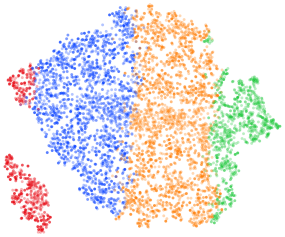


Fig. 9. t-SNE visualization of the pseudo-domain discriminative feature space.

results collectively verify the robustness and fidelity of our foundation-model-driven latent domain mining, providing a reliable basis for subsequent domain-aware prompt adaptation in multi-center postoperative scenarios.

F. Feature Decoupling and Long-tail Outlier Analysis

To intuitively validate the disentanglement capability of our framework, we visualized the latent feature space using t-SNE dimensionality reduction. As illustrated in Fig. 9, the feature embeddings spontaneously organize into four distinct, cohesive clusters. This topological structure confirms that LDM module effectively maps heterogeneous multi-center textures into separable distributions, establishing a robust foundation for the subsequent Adapter to retrieve precise domain-specific prompt priors.

TABLE VII
QUANTITATIVE EVALUATION ON LONG-TAIL OUTLIERS (42 CASES) CHARACTERIZED BY SEVERE MOTION ARTIFACTS AND EXTREME ANATOMICAL DEFORMATIONS.

Method	DSC (%) ↑	HD95 (mm) ↓	IoU (%) ↑
nnU-Net [17]	65.4	24.2	50.1
SurgicalSAM [22]	68.1	21.5	53.4
SAM-Med 3D (Baseline) [9]	61.2	28.5	46.8
DPD-SAM (Ours)	**79.4**	**10.2**	**66.5**
Improvement over Baseline	*+18.2*	*-18.3*	*+19.7*

Notably, the t-SNE visualization reveals a spatially isolated feature island in the lower-left region. Retrospective analysis identifies this cluster as long-tail outliers, specifically cases characterized by severe motion artifacts or extreme anatomical deformations. To quantitatively validate our framework's robustness against these edge cases, we isolated the 42 clinical cases within this cluster. As summarized in Table VII, the baseline SAM-Med 3D exhibits significant performance degradation on these challenging cases, achieving a DSC of only 61.2%. In contrast, DPD-SAM successfully maintains high precision. This confirms that our unsupervised clustering mechanism accurately captures and isolates pathological edge cases without requiring manual intervention. Unlike traditional methods where such outliers often disrupt global distribution learning, mapping extreme cases into a dedicated latent domain allows the prompt generator to synthesize specialized geometrical constraints, effectively mitigating the vulnerability of foundation models in extreme clinical scenarios.

G. Discussion

The proposed DPD-SAM framework establishes a robust baseline for post-operative glioma segmentation by synergizing latent domain mining with generative refinement. By generating dynamic, instance-specific prompts from unsupervised global embeddings, it overcomes multi-center heterogeneity without explicit labels, while the conditional diffusion module reduces boundary errors and helps recover small residual lesions missed by the initial pass, as evidenced by Fig. 5 and Table VII.

Despite these advances, limitations persist. With DDIM sampling using only 10 steps, the refinement adds only a bounded overhead on top of the SAM pass, giving a per-volume inference time of about 1.5 s on a single GPU, which is slightly slower than discriminative baselines yet far faster than a 1000-step sampler. As post-operative delineation is an offline analysis task that tolerates a minute-level latency rather than a real-time intra-operative one, this latency is well within practical requirements, and further acceleration is left for future work. Additionally, extreme motion artifacts may still challenge the prompt alignment in rare cases. Future work will focus on accelerating inference via progressive distillation and exploring robust contrastive strategies for better outlier handling. Furthermore, we plan to extend our framework to emerging advanced SAM foundation models, to leverage enhanced feature representations and incorporate latent-space contrastive learning for clinically deployable segmentation.

V. CONCLUSION

In this paper, we presented DPD-SAM, a unified framework designed to address the challenges of cross-center data heterogeneity and complex postoperative artifacts in glioma segmentation. By integrating unsupervised representation learning with generative refinement, our method effectively bridges the gap between foundation models and clinical specificities. Specifically, we introduced an LDM module that leverages the global semantic space of the SAM encoder to identify intrinsic imaging patterns without manual supervision. To efficiently adapt SAM to these discovered domains, we designed a Low-Rank DPG, which dynamically synthesizes instance-specific prompts to guide the generation of robust initial masks. Furthermore, we reformulated the boundary refinement task as a conditional generative process. The proposed CDR module, constrained by both geometric priors and domain styles, helps reduce small-residual omissions and boundary defects caused

by surgical cavities. Extensive experiments on four multi-center datasets demonstrated the superiority of our framework. Notably, DPD-SAM consistently outperformed state-of-the-art specialist models, particularly on the In-house dataset, where it reduced the HD95 to 6.9 mm, confirming its reliability in capturing precise surgical margins. These results highlight DPD-SAM's potential as a promising assistive tool for postoperative assessment, which nonetheless requires further prospective clinical validation before deployment, aligning with the RANO criteria for residual tumor quantification and treatment planning. By combining parameter-efficient adaptation of visual foundation models with conditional generative refinement, our framework offers new insights into handling heterogeneous medical imaging tasks.

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